

Compiler-Validated Semantic Editing for Weak Local Coding Models

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Abstract

Coding agents edit source code as text. Small locally runnable models fail at it. Diff formats break, identifiers get guessed, and errors surface only after files change. **ago** is a semantic edit protocol for Go. The agent queries a typechecked workspace and submits mutations from a versioned operation catalog. The compiler validates every mutation against an in-memory snapshot before anything reaches disk. Edits apply as atomic, generation-checked transactions or not at all. An edit that does not typecheck is unrepresentable, not merely checked. Rejections are the conversation. Each one carries typed diagnostics and paste-ready repairs, the corrected call written out verbatim. Resending a rejected call unchanged escalates. An oracle replays each task’s ground-truth commit through the protocol before any model attempts it, so a task enters the benchmark only once it is proven solvable. Tasks are mined from real commits in *traefik*, *vault*, and *boundary*. The evaluation is a grid of three modes. One edits files raw. One addresses symbols through an LSP with advisory validation (*Serena*). One sees only the protocol. All run against a frozen engine, so the benchmark cannot chase the tasks. Across three locally served models the mode ordering never inverts. The protocol beats symbol addressing, which beats raw text editing. The floor 9B model passes 41% of episodes through the protocol, against 32% with symbol addressing alone and 24% raw. The widest gap is the model that is a poor freeform tool-caller. Its raw pass rate collapses to 4%, yet recovers to 53% through the protocol. Sampled on-disk states during editing fail to compile 76% to 86% of the time in raw mode, and at most 2% in semantic mode, in every model. All code, per-episode evidence (transcripts, request logs, serving configurations), and analysis tooling are released.

1 Introduction

TODO(guy): expand from *idea.md*.

2 Related Work

Symbol-level agent tooling over LSP navigates semantically but still edits text, or exposes a single mutation. *Serena* [7] and *gopls*’s experimental MCP server [8] are examples. The compiler diagnostics sit right there in those stacks. The difference here is that validation is not advisory. CODESTRUCT [3] frames agents over structured action spaces with syntax-validated edits, and reports token and accuracy gains. **ago**’s mutations are type-checked, transactional, and carry repairs. Repository-scale editing benchmarks find instructed edits hard for current models

(RES-Q [4], FEA-Bench [5], EDIT-Bench [1]). Edit-format leaderboards [2] track a separate failure, where a model cannot emit a valid edit in the required format at all. That is the failure the repair channel converts. The format stays strict, and the rejection hands back the corrected call. Compiler-guided patch generation (Repilot [9]) and typed-hole context (Hazel [6]) take the same validate-during-generation stance, at expression granularity.

3 The ago Protocol

3.1 Surface

Ten tools carry a versioned catalog of 33 operations. They span declaration, statement, test, and project families. Seven query kinds return deterministic, paged responses. Node handles address statements inside declarations. Generation counters make staleness explicit instead of silent.

3.2 Transactions

An ordered op list validates as one unit against an in-memory snapshot. Ops apply to a copy. The dirty set re-typechecks once, in parallel, scheduled over post-edit import edges. Resolution proofs run, so a rename that would capture a reference is rejected even though the compiler would accept it. Then all files write together, or nothing does. Pre-existing issues in the affected packages are baselined and tolerated. Real codebases carry history, so only new diagnostics reject.

3.3 Rejection as conversation

Every rejection answers one question. What is the correct next call? It carries typed diagnostics, candidate lists, and `possible_repairs`, complete calls the agent resends verbatim. Repairs are tested by execution before they are offered. A repair that would itself reject is worse than none. Resending a rejected call unchanged escalates. Accepted mutations embed the fresh view of the touched declaration, so back-to-back edits skip the read round trip.

4 Oracle Methodology

Every benchmark task comes from a real commit, so the correct answer is already known. Before any model attempts a task, the oracle submits that known answer through the protocol itself. If the right answer cannot get through, the model never had a chance. The problem is then ours, whether an engine bug, an extractor error, or a named protocol ceiling. The task stays out of the roster until it is fixed. This separates “the model failed” from “the task was impossible” by construction. The oracle transcripts double as a training corpus.

5 Benchmark

Same model, same harness, same prompts, same time cap, three modes. Token parity across prompts is enforced by test. Raw edits files with shell tools. Serena addresses symbols through the language server, a pinned revision of the Serena MCP server, but its validation stays advisory. Diagnostics are available on request, and nothing blocks a broken write from landing. Semantic

sees only the protocol. The middle mode is the ablation that separates the two variables the protocol bundles, symbol addressing and mandatory validation. Scoring is a goal predicate, a clean typecheck, and scoped tests. The test gate is voided where a pristine worktree also fails it. Serving is pinned per run. GLM-4.7-Flash (MoE) runs at UD-Q4_K_XL, roughly 20 GiB, with a q8_0 K/V cache at 128k context. gpt-oss-20b runs at MXFP4, 11.28 GiB of weights before KV. Qwen3.5-9B runs at UD-Q4_K_XL, with its card’s embedding server evicted to fit the larger quant. Local serving is part of the experiment, constraints included. Every episode records its transcript, diff, score, token usage, failure kind, and the daemon’s request log. The evidence is the contribution as much as the numbers.

5.1 Evaluation freeze

Every run before the freeze tag is development data. Rejections became repairs, and oracle failures became engine fixes. That loop is a contribution, but it cannot also be the headline evidence. The engine is frozen at a tagged revision (`eval-freeze-1`). Every episode records the engine revision it ran under, so the split is auditable. Headline tables come only from frozen-tag runs over the development set of 20 oracle-certified tasks. A held-out set mined from etcd, caddy, and hugo is extracted and certified only after the tag, with the frozen engine. A certification failure there is reported as a protocol ceiling, not fixed.

5.2 Invalid intermediate states

The mechanism claim is that validation before disk changes what a repository passes through, not just where an episode ends. A watcher samples each worktree during the agent’s editing window. When the mtime fingerprint of the Go files changes and then holds still for one interval, the settled state gets a `go build ./...` verdict. The oracle runs as the control and measures the sampler itself. Raw and advisory modes pass through whatever the model writes. The semantic mode cannot land a declaration that does not typecheck, so its measured rate is the write window itself. A mutation touching dozens of call sites writes files sequentially, and a sample can catch the half-written state. Replaying every accepted mutation from the flagged episodes shows each settled state compiles. No host tool wrote Go in any of them.

6 Results

Pooled over the frozen grid, semantic passed 173/340 episodes. Raw passed 78/340. The per-task table tells the sharper story. Raw wins only where grep suffices, and goes to zero on the tasks that need whole-workspace consistency.

Two readings stand out. First, the mode ordering (Figure 1) never inverts across models. Symbol addressing helps, and mandatory validation helps again on top of it. Second, the protocol’s advantage is not uniform. It is widest for the model that edits text worst. A strong instruction-follower that is nonetheless a poor freeform tool user passes almost nothing raw, yet recovers to a majority through the protocol. The protocol removes exactly the failure it is prone to. The invalid-intermediate rate (Figure 2) is the mechanism. Both text modes leave the workspace uncompileable most of the time mid-edit, while the semantic mode holds it valid throughout.

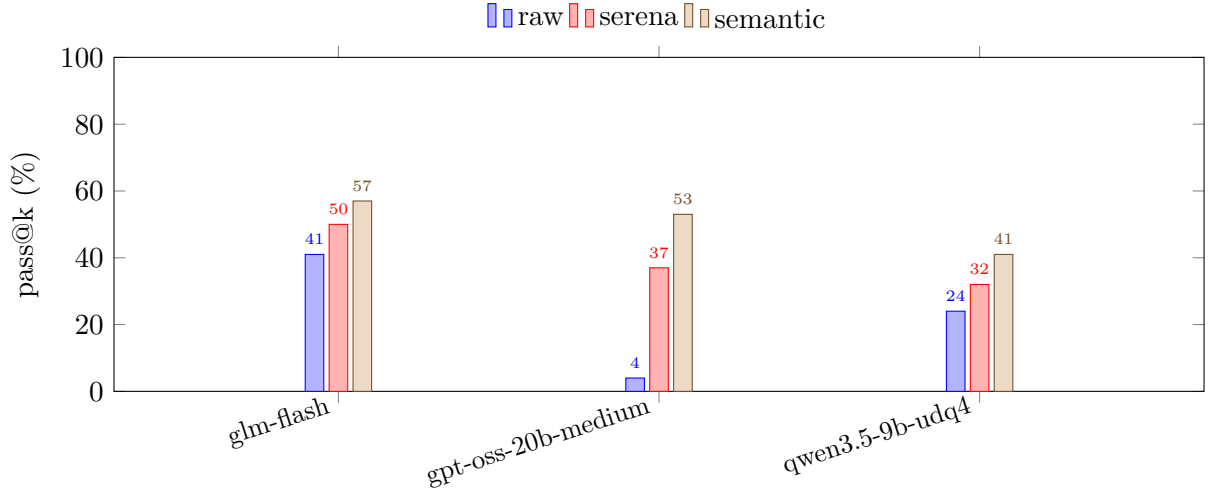


Figure 1: Pass@k by mode, per model. The ordering $\text{semantic} \geq \text{serena} \geq \text{raw}$ holds for every model in the grid. The lift is largest where the model is a weak freeform tool-caller (gpt-oss-20b-medium at 4% raw, 53% semantic). Generated from committed episode evidence.

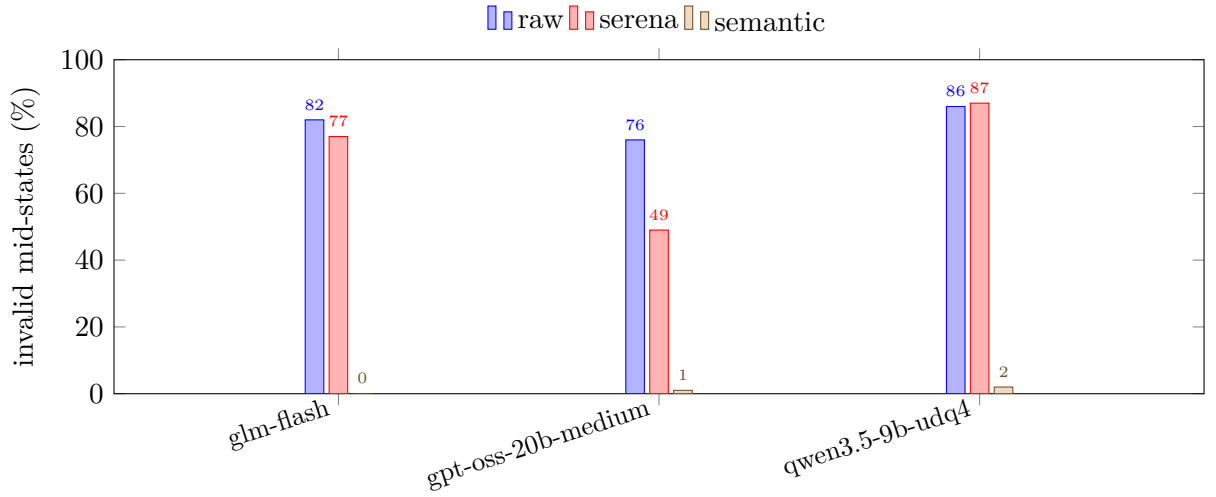


Figure 2: Fraction of sampled on-disk states that fail `go build` during the editing window. The text-editing modes churn through broken states. The semantic mode cannot land a declaration that does not typecheck, so its residual is only the multi-file write window. Generated from the same evidence.

Table 1: Per-task, per-mode outcomes across the frozen grid. Generated from committed episode evidence. Each pass@k is over k episodes. Median time-to-green is over passing episodes.

profile	task	mode	pass@k	median green (s)	capped	resends
glm-flash	boundary_06d69f69	raw	6/6	244	0	0
glm-flash	boundary_06d69f69	semantic	6/6	250	0	2
glm-flash	boundary_06d69f69	serena	6/6	178	0	0
glm-flash	boundary_35b91c66	raw	2/6	544	4	0
glm-flash	boundary_35b91c66	semantic	3/6	379	3	12
glm-flash	boundary_35b91c66	serena	5/6	247	0	0
glm-flash	boundary_57ffc718	raw	0/6	–	6	0
glm-flash	boundary_57ffc718	semantic	2/6	469	4	29
glm-flash	boundary_57ffc718	serena	0/6	–	6	0
glm-flash	boundary_687dd1bd	raw	0/6	–	3	0
glm-flash	boundary_687dd1bd	semantic	0/6	–	6	72
glm-flash	boundary_687dd1bd	serena	0/6	–	5	0
glm-flash	boundary_9237d6f7	raw	1/6	554	2	0
glm-flash	boundary_9237d6f7	semantic	3/6	401	2	98
glm-flash	boundary_9237d6f7	serena	5/6	175	0	0
glm-flash	boundary_9354a4eb	raw	0/6	–	6	0
glm-flash	boundary_9354a4eb	semantic	0/6	–	6	11
glm-flash	boundary_9354a4eb	serena	0/6	–	6	0
glm-flash	boundary_9f2c83f3	raw	2/6	671	5	0
glm-flash	boundary_9f2c83f3	semantic	5/6	407	0	6
glm-flash	boundary_9f2c83f3	serena	3/6	675	0	0
glm-flash	boundary_b26814a3	raw	1/6	720	5	0
glm-flash	boundary_b26814a3	semantic	4/6	669	4	30
glm-flash	boundary_b26814a3	serena	6/6	406	0	0
glm-flash	boundary_b4b95e0f	raw	0/6	–	6	0
glm-flash	boundary_b4b95e0f	semantic	0/6	–	6	7
glm-flash	boundary_b4b95e0f	serena	0/6	–	6	0
glm-flash	boundary_bf1486f7	raw	0/6	–	6	0
glm-flash	boundary_bf1486f7	semantic	4/6	657	4	8
glm-flash	boundary_bf1486f7	serena	0/6	–	6	0
glm-flash	boundary_c0532428	raw	6/6	453	1	0
glm-flash	boundary_c0532428	semantic	6/6	334	0	13
glm-flash	boundary_c0532428	serena	6/6	138	0	0
glm-flash	boundary_caf19f86	raw	6/6	715	3	0
glm-flash	boundary_caf19f86	semantic	6/6	325	0	6
glm-flash	boundary_caf19f86	serena	6/6	171	0	0
glm-flash	boundary_dd2c3807	raw	1/6	900	3	0
glm-flash	boundary_dd2c3807	semantic	3/6	720	6	21
glm-flash	boundary_dd2c3807	serena	1/6	741	4	0
glm-flash	boundary_eb61ac63	raw	0/6	–	6	0
glm-flash	boundary_eb61ac63	semantic	4/6	720	6	21
glm-flash	boundary_eb61ac63	serena	0/6	–	6	0
glm-flash	boundary_fb1ea92a	raw	6/6	431	0	0
glm-flash	boundary_fb1ea92a	semantic	0/6	–	6	81
glm-flash	boundary_fb1ea92a	serena	2/6	323	2	0
glm-flash	traefik_51491463	raw	5/6	272	0	0
glm-flash	traefik_51491463	semantic	6/6	268	0	3
glm-flash	traefik_51491463	serena	5/6	128	0	0
glm-flash	vault_7103bc2c	raw	0/6	–	3	0
glm-flash	vault_7103bc2c	semantic	0/6	–	6	20

profile	task	mode	pass@k	median green (s)	capped	resends
glm-flash	vault_7103bc2c	serena	0/6	–	5	0
glm-flash	vault_95b24f55	raw	6/6	340	0	0
glm-flash	vault_95b24f55	semantic	6/6	324	0	4
glm-flash	vault_95b24f55	serena	6/6	248	0	0
glm-flash	vault_caec65a7	raw	1/6	720	6	0
glm-flash	vault_caec65a7	semantic	6/6	425	0	0
glm-flash	vault_caec65a7	serena	5/6	456	2	0
glm-flash	vault_cfff8d42	raw	6/6	346	0	0
glm-flash	vault_cfff8d42	semantic	4/6	720	6	26
glm-flash	vault_cfff8d42	serena	4/6	299	3	0
gpt-oss-20b-medium	boundary_06d69f69	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_06d69f69	semantic	6/6	153	0	1
gpt-oss-20b-medium	boundary_06d69f69	serena	6/6	142	0	0
gpt-oss-20b-medium	boundary_35b91c66	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_35b91c66	semantic	6/6	311	0	5
gpt-oss-20b-medium	boundary_35b91c66	serena	6/6	124	0	0
gpt-oss-20b-medium	boundary_57ffc718	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_57ffc718	semantic	1/6	231	0	9
gpt-oss-20b-medium	boundary_57ffc718	serena	0/6	–	0	0
gpt-oss-20b-medium	boundary_687dd1bd	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_687dd1bd	semantic	0/6	–	0	9
gpt-oss-20b-medium	boundary_687dd1bd	serena	0/6	–	0	0
gpt-oss-20b-medium	boundary_9237d6f7	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_9237d6f7	semantic	5/6	304	0	7
gpt-oss-20b-medium	boundary_9237d6f7	serena	5/6	90	0	0
gpt-oss-20b-medium	boundary_9354a4eb	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_9354a4eb	semantic	0/6	–	0	7
gpt-oss-20b-medium	boundary_9354a4eb	serena	0/6	–	0	0
gpt-oss-20b-medium	boundary_9f2c83f3	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_9f2c83f3	semantic	5/6	278	0	2
gpt-oss-20b-medium	boundary_9f2c83f3	serena	2/6	135	0	0
gpt-oss-20b-medium	boundary_b26814a3	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_b26814a3	semantic	0/6	–	0	7
gpt-oss-20b-medium	boundary_b26814a3	serena	0/6	–	0	0
gpt-oss-20b-medium	boundary_b4b95e0f	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_b4b95e0f	semantic	0/6	–	0	6
gpt-oss-20b-medium	boundary_b4b95e0f	serena	0/6	–	0	0
gpt-oss-20b-medium	boundary_bf1486f7	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_bf1486f7	semantic	3/6	160	0	10
gpt-oss-20b-medium	boundary_bf1486f7	serena	0/6	–	0	0
gpt-oss-20b-medium	boundary_c0532428	raw	1/6	124	0	0
gpt-oss-20b-medium	boundary_c0532428	semantic	6/6	130	0	5
gpt-oss-20b-medium	boundary_c0532428	serena	2/6	141	0	0
gpt-oss-20b-medium	boundary_caf19f86	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_caf19f86	semantic	6/6	242	0	0
gpt-oss-20b-medium	boundary_caf19f86	serena	6/6	100	0	0
gpt-oss-20b-medium	boundary_dd2c3807	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_dd2c3807	semantic	0/6	–	0	11
gpt-oss-20b-medium	boundary_dd2c3807	serena	0/6	–	0	0
gpt-oss-20b-medium	boundary_eb61ac63	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_eb61ac63	semantic	3/6	133	0	1
gpt-oss-20b-medium	boundary_eb61ac63	serena	0/6	–	0	0
gpt-oss-20b-medium	boundary_fb1ea92a	raw	0/6	–	0	0
gpt-oss-20b-medium	boundary_fb1ea92a	semantic	0/6	–	0	20
gpt-oss-20b-medium	boundary_fb1ea92a	serena	0/6	–	0	0
gpt-oss-20b-medium	traefik_51491463	raw	0/6	–	0	0

profile	task	mode	pass@k	median green (s)	capped	resends
gpt-oss-20b-medium	traefik_51491463	semantic	5/6	176	0	8
gpt-oss-20b-medium	traefik_51491463	serena	2/6	87	0	0
gpt-oss-20b-medium	vault_7103bc2c	raw	0/6	—	0	0
gpt-oss-20b-medium	vault_7103bc2c	semantic	0/6	—	0	7
gpt-oss-20b-medium	vault_7103bc2c	serena	0/6	—	0	0
gpt-oss-20b-medium	vault_95b24f55	raw	0/6	—	0	0
gpt-oss-20b-medium	vault_95b24f55	semantic	6/6	351	0	2
gpt-oss-20b-medium	vault_95b24f55	serena	6/6	177	0	0
gpt-oss-20b-medium	vault_caec65a7	raw	1/6	159	0	0
gpt-oss-20b-medium	vault_caec65a7	semantic	6/6	350	0	0
gpt-oss-20b-medium	vault_caec65a7	serena	4/6	256	0	0
gpt-oss-20b-medium	vault_cff8d42	raw	3/6	116	0	0
gpt-oss-20b-medium	vault_cff8d42	semantic	6/6	115	0	9
gpt-oss-20b-medium	vault_cff8d42	serena	5/6	204	0	0
qwen3.5-9b-udq4	boundary_06d69f69	raw	5/5	61	0	0
qwen3.5-9b-udq4	boundary_06d69f69	semantic	5/5	169	0	0
qwen3.5-9b-udq4	boundary_06d69f69	serena	5/5	92	0	0
qwen3.5-9b-udq4	boundary_35b91c66	raw	0/5	—	0	0
qwen3.5-9b-udq4	boundary_35b91c66	semantic	3/5	275	0	2
qwen3.5-9b-udq4	boundary_35b91c66	serena	3/5	142	0	0
qwen3.5-9b-udq4	boundary_57ffc718	raw	0/5	—	0	0
qwen3.5-9b-udq4	boundary_57ffc718	semantic	1/5	204	0	3
qwen3.5-9b-udq4	boundary_57ffc718	serena	0/5	—	1	0
qwen3.5-9b-udq4	boundary_687dd1bd	raw	1/5	217	0	0
qwen3.5-9b-udq4	boundary_687dd1bd	semantic	0/5	—	0	2
qwen3.5-9b-udq4	boundary_687dd1bd	serena	0/5	—	0	0
qwen3.5-9b-udq4	boundary_9237d6f7	raw	0/5	—	0	0
qwen3.5-9b-udq4	boundary_9237d6f7	semantic	1/5	676	0	2
qwen3.5-9b-udq4	boundary_9237d6f7	serena	3/5	66	0	0
qwen3.5-9b-udq4	boundary_9354a4eb	raw	0/5	—	0	0
qwen3.5-9b-udq4	boundary_9354a4eb	semantic	0/5	—	0	4
qwen3.5-9b-udq4	boundary_9354a4eb	serena	0/5	—	0	0
qwen3.5-9b-udq4	boundary_9f2c83f3	raw	0/5	—	0	0
qwen3.5-9b-udq4	boundary_9f2c83f3	semantic	3/5	134	0	3
qwen3.5-9b-udq4	boundary_9f2c83f3	serena	3/5	136	1	0
qwen3.5-9b-udq4	boundary_b26814a3	raw	0/5	—	0	0
qwen3.5-9b-udq4	boundary_b26814a3	semantic	2/5	204	0	10
qwen3.5-9b-udq4	boundary_b26814a3	serena	0/5	—	1	0
qwen3.5-9b-udq4	boundary_b4b95e0f	raw	0/5	—	1	0
qwen3.5-9b-udq4	boundary_b4b95e0f	semantic	0/5	—	0	0
qwen3.5-9b-udq4	boundary_b4b95e0f	serena	0/5	—	0	0
qwen3.5-9b-udq4	boundary_bf1486f7	raw	0/5	—	0	0
qwen3.5-9b-udq4	boundary_bf1486f7	semantic	1/5	122	0	1
qwen3.5-9b-udq4	boundary_bf1486f7	serena	0/5	—	0	0
qwen3.5-9b-udq4	boundary_c0532428	raw	5/5	132	0	0
qwen3.5-9b-udq4	boundary_c0532428	semantic	3/5	204	0	17
qwen3.5-9b-udq4	boundary_c0532428	serena	3/5	92	0	0
qwen3.5-9b-udq4	boundary_caf19f86	raw	3/5	328	0	0
qwen3.5-9b-udq4	boundary_caf19f86	semantic	1/5	50	0	5
qwen3.5-9b-udq4	boundary_caf19f86	serena	4/5	98	0	0
qwen3.5-9b-udq4	boundary_dd2c3807	raw	0/5	—	0	0
qwen3.5-9b-udq4	boundary_dd2c3807	semantic	3/5	166	0	49
qwen3.5-9b-udq4	boundary_dd2c3807	serena	0/5	—	0	0
qwen3.5-9b-udq4	boundary_eb61ac63	raw	0/5	—	0	0
qwen3.5-9b-udq4	boundary_eb61ac63	semantic	0/5	—	0	15
qwen3.5-9b-udq4	boundary_eb61ac63	serena	0/5	—	0	0

profile	task	mode	pass@k	median green (s)	capped	resends
qwen3.5-9b-udq4	boundary_fb1ea92a	raw	0/5	–	0	0
qwen3.5-9b-udq4	boundary_fb1ea92a	semantic	0/5	–	0	4
qwen3.5-9b-udq4	boundary_fb1ea92a	serena	0/5	–	1	0
qwen3.5-9b-udq4	traefik_51491463	raw	4/5	85	0	0
qwen3.5-9b-udq4	traefik_51491463	semantic	5/5	136	0	1
qwen3.5-9b-udq4	traefik_51491463	serena	3/5	111	0	0
qwen3.5-9b-udq4	vault_7103bc2c	raw	0/5	–	0	0
qwen3.5-9b-udq4	vault_7103bc2c	semantic	0/5	–	0	3
qwen3.5-9b-udq4	vault_7103bc2c	serena	0/5	–	1	0
qwen3.5-9b-udq4	vault_95b24f55	raw	1/5	98	0	0
qwen3.5-9b-udq4	vault_95b24f55	semantic	5/5	274	0	0
qwen3.5-9b-udq4	vault_95b24f55	serena	4/5	228	0	0
qwen3.5-9b-udq4	vault_caec65a7	raw	0/5	–	1	0
qwen3.5-9b-udq4	vault_caec65a7	semantic	5/5	590	1	0
qwen3.5-9b-udq4	vault_caec65a7	serena	2/5	286	1	0
qwen3.5-9b-udq4	vault_cfff8d42	raw	5/5	60	0	0
qwen3.5-9b-udq4	vault_cfff8d42	semantic	3/5	291	1	10
qwen3.5-9b-udq4	vault_cfff8d42	serena	2/5	257	0	0

7 Limitations

One language. Contamination is held constant across modes but inflates absolute rates. One serving stack per model, small k, and the infrastructure is author-run. TODO: expand honestly.

8 Availability

Code (MIT), all episode evidence, and analysis tooling are at <https://github.com/guygrigsby/agent-go>.

References

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